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Effects of chronic experimental N deposition on northern hardwood forests fade after cessation of N additions

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Abstract

Background Anthropogenic nitrogen (N) deposition has the potential to affect forest ecosystem carbon (C) storage and sequestration both positively and negatively. Effects on aboveground biomass increment seem to be largely positive, but the impact of elevated N inputs on soil microbial systems in forests is often negative, reducing microbial biomass and soil respiration rates. Twenty-four years of experimental N deposition at a rate of 3 g N m^{-2} y^{-1} broadly suppressed soil microorganisms across four northern hardwood forest sites. To assess responses to reductions in chronic N inputs, soil and sugar maple root samples from the study sites were collected and examined for phospholipid fatty acid (PLFA) and neutral lipid fatty acid (NLFA) indicators of microbial groups at one- and three-years after the cessation of the N deposition treatment, and tree growth was measured annually for six years after treatments stopped.

Results In the first year post-treatment, all soil and root microbial indicators remained suppressed, but by three years post-cessation of N additions, indicators for soil fungi and arbuscular mycorrhizal fungi (AMF) showed much less suppression, with several bacterial groups and measures of the total microbial community showing full recovery. By year three, significant differences also no longer existed between control and previously N-amended plots for PLFA and NLFA biomarkers indicative of AMF in tree fine roots. During active N loading, trees receiving the N deposition treatment increased their annual aboveground growth increment, in concert with reduced carbon allocation to AMF, but this growth enhancement faded during the six years after cessation of N additions.

Conclusions Indicators of suppressed microbial community abundance showed gradual recovery from one to three years after cessation of long-term experimental N loading. This recovery indicates the potential for microbially-mediated ecosystem C sequestration responses to high anthropogenic N deposition, such as enhanced tree growth, reduced decomposition rates, and increased surface soil carbon content, to also fade when N inputs are reduced.

Keywords Microbial community, Arbuscular mycorrhizal fungi, Tree growth, Carbon sequestration, Carbon storage

Introduction

Nitrogen (N) is an essential nutrient in the biosphere, the lack of which frequently limits plant growth. Biological N fixation by microorganisms is the single largest biogenic input of N to soil globally. However, anthropogenic sources of reactive N, tallied together, exceed biological sources (Galloway and Cowling 2002). These anthropogenic N sources can lead to large N inputs to ecosystems that were historically N limited (Fleischer et al. 2013; Matson et al. 1999; Meuiner et al. 2016).

In regions of high N emissions, including heavily industrialized and heavily farmed areas, atmospheric N deposition rates are much higher than in areas of low N emissions (Zhu et al. 2015). Heightened levels of N inputs can be a benefit to forest systems, as more N can allow for more plant growth (Helfield and Naiman 2001; Raison

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and Myers 1992; Schulte-Uebbing et al. 2022). In the four sugar maple-dominated forest ecosystems in Michigan where this project was conducted, annual aboveground net primary productivity (ANPP) increased significantly in response to twenty-four years of experimental N deposition additions of 3 g N m^{-2} y^{-1} (Pregitzer et al. 2008), due to increases in aboveground woody biomass increment. The increases in aboveground tree growth at the sites did not stem from increases in photosynthesis (Talhelm et al. 2011). Instead, it was concluded that enhanced aboveground growth was due, at least in part, to reduced belowground C allocation. Other studies have also observed this mechanism, usually due to reduced root biomass (Frey et al. 2014; Magill et al. 2000). However, in the four Michigan sites the response was not due to reduced C allocation to root biomass and respiration, which were not changed by the experimental N deposition (Burton et al. 2004). Rather, enhanced aboveground growth was attributed to significant decreases in the abundance of arbuscular mycorrhizal fungi (AMF) (van Diepen et al. 2007, 2010, 2011). The concurrent trends of increased tree growth and decreased AMF suggested that these fungal networks received less C from their tree hosts under elevated N deposition, leaving more C to be allocated for other uses, including aboveground tree growth (Talhelm et al. 2011).

For soil bacterial and fungal communities in forests, elevated N inputs often decrease biodiversity, abundance, and activity (Canfield et al. 2010; van Diepen et al. 2010). Effects of the changes in soil microbial communities are variable. For example, in the northern hardwood forests used in this study, soil microbial community function and composition shifted under experimentally elevated N deposition (Maaroufi et al. 2015; Tian et al. 2017; Zak et al. 2011), with broad suppression of many microbial groups (Van Diepen et al. 2010). Populations of actinobacteria were reduced in top soil layers (Eisenlord and Zak 2010), as were those of ligninolytic fungi (Entwhistle et al. 2018). An increase in C storage arose from depressed decomposition rates and lowered expression of ligninolytic enzymes (Carreiro et al. 2000; DeForest et al. 2004; Edwards et al. 2011; Frey et al. 2014; Pregitzer et al. 2008; Zak et al. 2008). Further, chronic N additions were found to decrease soil respiration, and this decrease was attributed to decreases in soil microbial respiration (Burton et al. 2004; Frey et al. 2014; Liu and Greaver 2010).

Although impacts on the microbial community, decomposition processes and soil C have been demonstrated in response to chronic elevated N inputs, the degree to which these changes might persist or fade if N inputs decline is not well investigated. Such knowledge is needed, as rates and patterns of global N deposition are changing, with some regions seeing significant reductions in N inputs while others have increases (Butler et al. 2005; Chang et al. 2022; Liu et al. 2022).

To begin addressing this issue, we analyzed phospholipid fatty acids (PLFAs) and neutral lipid fatty acids (NLFAs) from samples of surface soil and fine roots during the first and third years after the cessation of long-term experimental N deposition (3 g N m^{-2} y^{-1} for 24 years) to four northern hardwood forests in Michigan. PLFAs are a primary component of cell membranes. The specific form of these acids, with regards to saturated vs unsaturated and branched vs unbranched tails and side chains, can indicate the biological group from which they originate. Because they rapidly decay in soil, measurements of PLFA can be assumed to originate from living cells (Frostegård et al. 2011). In eukaryotes, NLFA can be used to store energy (Bååth 2003). Individually, both metrics give a snapshot of general abundance on the level of microbial groups. The 16:1 ω5c biomarkers for PLFA and NLFA are associated with AMF. In fine roots, this PLFA indicates AMF biomass while the NLFA indicates fungal C storage within the root. Coupling them together gives a more complete picture of AMF responses specifically and helps differentiate the biomarker from bacteria (Ngosong et al. 2012). Soil concentrations of PLFA and NLFA 16:1 ω5c are indicative of extraradical AMF hyphal biomass.

Our objectives were to determine: 1) the degree to which the soil microbial community, and fungi in particular, continued to be suppressed, consistent with responses throughout the period of N addition; and 2) if tree growth continued to be enhanced following the cessation of the experimental N deposition treatment. In addition to microbial members of the decomposer community, the abundance of AMF in both roots and soil was of interest due to the role AMF reduction appeared to play in the enhanced aboveground growth observed during the N addition period. Soil fungi in multiple guilds help determine ecosystem C storage via decomposition rates and via aboveground biomass accumulation. By studying the community as a whole, we were able to track both positive and negative impacts that might be related to overall forest C storage. Given the very long duration of N additions and potential for greatly enhanced site N capital as a result, we hypothesized that at one- and three-years post-cessation, soil microbial communities would broadly show continued suppression in formerly treated plots as compared to control plots. Specifically, we expected to see continued enhancement of aboveground tree diameter growth, matching a continued reduction in AMF abundance in soils and roots. Additionally, we hypothesized that soil fungi would remain suppressed and that their suppression would maintain

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the build-up of litter material in forest floors and top soil layers.

Methods

Study sites

This study was conducted along the Michigan Gradient, which consists of four northern hardwood sites spanning a 500 km latitudinal gradient in Michigan, USA (Fig. 1). All sites are dominated by sugar maple (*Acer saccharum* Marsh.) and have similar loamy sand soils (Burton et al. 1991, Table 1). Each site has six 30 m × 30 m plots, of which three were left untreated as controls and three received experimental N deposition from 1994 to 2017.

To simulate chronic elevated N deposition at the sites, sodium nitrate pellets were spread at a rate of 3 g N m^{-2} y^{-1}, in six 0.5 g N m^{-2} increments spread every four weeks across the growing season (early May to mid-September). This occurred for 24 years starting in 1994. Active N loading ended in the fall of 2017. Since then, plots have been allowed to recover.

Soil and root sampling

Most soils and roots used in this study were collected in August 2020 from all four sites. In addition, samples were collected in August 2018 from Site A. For sampling, each plot was subdivided into six 10 m × 15 m sub-plots. Samples were taken at random locations within a sub-plot and consisted of paired 2 cm diameter, 10 cm deep cores of decomposed forest floor horizons (Oe/Oa) and surface mineral soil, taken beneath the loose litter layer (Oi). One of each pair was designated for soil analysis and one for root analysis, and the six root or soil samples per plot were composited in the field, creating one root and one soil sample per plot. All samples were frozen until further analysis could be conducted and freeze-dried immediately before processing to ensure the loss of all moisture.

Soil samples were sifted through 2 mm and 1 mm screens successively, removing all rocks, roots, and coarse organic material. The resulting fine soil material was then used for the soil PLFA/NLFA analysis. Sugar maple roots were collected and sorted from the samples designated for root analysis. Sugar maple roots, distinctive enough to be separated by hand (Goldfarb et al. 1990), were sorted from the surrounding soil, which then was also put through 2 mm and 1 mm screens to ensure all root material was found. Roots were sorted to obtain those with a diameter less than 0.5 mm, and these were then cleaned via sonication for five minutes and finely minced by hand via scissors. The resulting root pieces were processed in the same manner as the soil for PLFA/NLFA. Microbial markers for AMF, in which we were

Table 1 Site characteristics of the four northern hardwood forests in Michigan, USA in 2018, after the end of the 24-year N deposition treatment period

Characteristic	Site A	Site B	Site C	Site D
Latitude (N)	46° 52′	45° 33′	44° 23′	43° 40′
Longitude (W)	88° 53′	84° 51′	85° 50′	86° 09′
Basal area ($m^2 ha^{-1}$)	35.7	33.6	33.9	37.7
Sugar maple (% of basal area)	91	86	79	71
Mean annual precipitation[a] (mm)	879	874	913	824
Growing season precipitation[a] (mm)	401	388	393	379
Mean annual temperature[b] (°C)	4.9	6.2	7.0	7.7
Growing season temperature[b] (°C)	15.0	16.0	16.2	16.8
Growing season length[c] (days)	138	154	160	163
Overstory age in 2018 (y)	111	105	106	110
Wet + dry total N deposition (g N m^{-2} y^{-1})[d]	0.50	0.60	0.75	0.90

[a] Mean annual and growing season (May–September) precipitation was recorded using weighing rain gages (Model 5-780, Belfort Instrument Co., Baltimore, MD) located in open areas within 5 km of each site

[b] Mean annual and growing season (May–September) temperature was recorded on site at 2 m using thermistors which were read every 30 min throughout the year, with averages recorded every 3 h using data loggers (EasyLogger Models 824 and 925, Data Loggers, Inc., Logan UT)

[c] Growing season length, for the years 2010 to 2019, was determined annually by project personnel, based on visual estimates of > 50% leaf expansion in the spring and < 75% of foliage remaining in the fall

[d] Estimated from CASTNET (Clean Air Status and Trends Network) total N deposition maps for the period 2017–2019 (https://www3.epa.gov/castnet/totaldep.html, accessed February 24, 2025)



Fig. 1 Locations of the four sites that comprise the Michigan Gradient study. All sites consist of a sugar maple dominated overstory growing on loamy sand soils

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particularly interested, can be found both inside sugar maple roots and in the surrounding soil.

PLFA and NLFA analysis

Methods used in this study follow those used in earlier PLFA/NLFA studies of the soil microbial communities in these sites, allowing us to compare our results to previous work and assemble a time-step analysis of changes in treatment effects on soil organisms after the cessation of the N addition treatments, relative to effects documented during the period of active N addition. The methods used are detailed in Frostegård et al. (1991), a modification of the Bligh & Dyer (1959) method. Phospholipid fatty acid (PLFA) analysis was performed on the freeze-dried soils and ground root samples independently. Soil samples analyzed were 2.5 g ± 0.3 g, and root samples were to 15 mg ± 6 mg.

Lipids were first extracted from the samples using 4 mL of phosphate buffer, 10 mL of methanol, and 5 mL of chloroform, vortexed and sonicated, and left to separate. Samples were then centrifuged at 2000 rpm for ten minutes, with the organic lipids concentrated in the chloroform supernatant and collected. An additional 5 mL of chloroform was added, and the process repeated. 5 mL of nanopure water was added to each sample, swirled by hand, and left to sit overnight, after which the organic lipids were subjected to silicic acid chromatography with 2.5 mL of chloroform, 2.5 mL of acetone twice, and with 2.5 mL of methanol four times. The neutral and phospholipid fractions were evaporated with the N-EVAP at 37 °C. The neutral and polar fractions were further processed by lipid methylation (Miller et al. 2004), creating fatty acid methyl esters (FAMEs), with 0.5 mL each of chloroform and methanol added to each sample. Then 1 mL of methanolic KOH was added to each sample and mixed. To each sample, 2 mL of hexane and 200 µL of 1 M acetic acid were added, triggering phase separation. 2 mL of nanopure water were added to break phase. Samples were vortexed for 30 s and then centrifuged for five minutes at 2000 rpm. This separated the samples into a bottom aqueous layer and a top hexane layer containing the FAMEs. The procedure was repeated two more times, and the hexane from all three extractions was evaporated in the N-EVAP at 37 °C in the dark.

A 19:0 standard was prepared to a concentration of 20 ng μL^{-1}. This standard is a known fatty acid not found in nature, dissolved in hexane. To each neutral acid sample, 500 µL of the standard were added. Phospholipid samples were dissolved in 250 µL of the standard. Samples were analyzed via gas chromatography (Agilent Technologies 6850 Network GC System, Agilent Technologies Inc., Santa Clara, CA). Peaks were matched to their corresponding microbial groups using their individual retention times as obtained from Sherlock™ Microbial Identification system software (MIDI Inc., Newark, Delaware). Peak areas were converted from their unitless original form to nmol g^{-1} or nmol mg^{-1} for soils and roots, respectively, based on initial sample mass extracted, dilutions occurring during processing, and the internal standard.

Tree growth

All trees greater than 5 cm in diameter at breast height (dbh) have permanent markings at 1.37 m aboveground, at which dbh was measured annually during the dormant season. Based on these diameter measurements, the annual basal area growth for each tree was calculated as the increase in tree cross-sectional area from one year to the next. Growth increments for all trees on each 30 m × 30 m plot were then summed to estimate plot level basal area increment (m^2 ha^{-1}) for each year from 1994 through 2023.

Statistical analysis

Statistical analyses were performed in R Studio (R Core team 2018). Simple analysis of variance (ANOVA) examined the effects of treatment for the samples from Site A collected in 2018. Two-factor fixed-effect ANOVA models were used to assess the effects of site and treatment (control vs former N deposition) on FAME concentrations for individual PLFA and NLFA markers in roots and soil for the samples collected in 2020. Since differences among sites were expected, site was included as a factor in the ANOVA to allow greater statistical power for assessing treatment effects. Treatment effects for basal area increment were assessed using a repeated measures (year) two-factor (study site, treatment) analysis of variance, with the years of simulated atmospheric N deposition (1994 to 2017) analyzed separately from the years following the cessation of N additions (2018 to 2023).

Results

Microbial indicators

At Site A, one year after the cessation of the N deposition treatment, concentrations of PLFA and NLFA 16:1 ω5c biomarkers associated with AMF in fine root tissue were reduced by 35 and 61%, respectively, for the former N deposition treatment relative to the control (Fig. 2). Soil concentrations of PLFA and NLFA 16:1 ω5c also indicated a reduction in extraradical AMF biomass (Figs. 3 and 4). In Site A, one year after discontinuing N additions, concentrations of all soil PLFA and NLFA microbial biomarkers were lower for the former N deposition treatment than for the control, nine of fifteen significantly so, indicating an overall reduction in fungal and bacterial biomass (Figs. 3 and 4).

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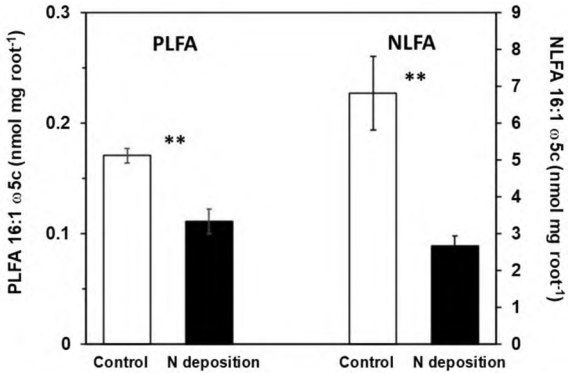


Fig. 2 Concentrations of PLFA and NLFA biomarkers for AMF within fine roots at Site A in 2018. Significant differences between the control and former N deposition treatment for individual biomarkers are indicated by ** for $P < 0.05$. Error bars are one standard error of the mean for three plots per treatment

Three years after the cessation of N additions, an examination of PLFA and NLFA in soils from all four study sites found fewer detectable differences between control plots and former N deposition plots for microbial biomarkers (Table 2, Table S1 in Supplemental Material). PLFA markers for AMF, bacteria, and fungi and an NFLA marker for fungi were significantly lower in samples from former N deposition plots than for the control (Table 2, Fig. S1). A measure of total NLFA, however, showed higher levels in the former treated plots (Fig. S2). All other microbial group indicators showed no significant differences between the former nitrogen deposition plots and the controls (Table 2, Table S1). For root samples, the only significant differences found were among sites, with no effects of the former N deposition treatment occurring (Table 2).

Two-way ANOVA tests examined potential interactions between site and treatment in affecting PLFA and NLFA levels in soil. Only three of these tests returned significant *P*-values for interactions, all of which were for PLFAs (Table 2). Two general measures of bacteria and the measure of gram-positive bacteria were significantly different for the interaction (*P*-values 0.045, 0.021, and 0.080, respectively). The interaction between site and treatment was not significant for any microbial group found in roots for either PLFA or NLFA (Table 2).

Tree growth

During the period of active N deposition treatment, aboveground tree growth was greater for treated plots than the control (Fig. 5), with the actual relationship being complex. Significant year by treatment, year by site, and year by treatment by site interactions were detectable throughout the time span (Table S1, Supplemental Material). These interactions are illustrated by greater growth at more southern sites (Fig. 6), stronger treatment effects in some years than others (Fig. 5), and a lack of treatment effects at Site D, in contrast to effects at the other three sites (Fig. 6). After cessation of the N deposition treatment, enhancement of tree growth on the former N deposition plots slowed over time (Fig. 5, with the differences between the control and N deposition plots no longer being significant overall (Table S2 in Supplemental Material).

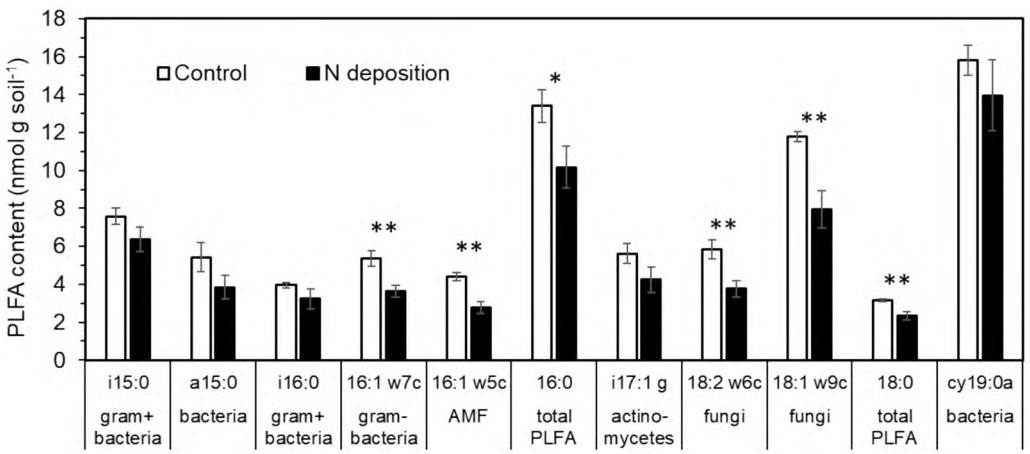


Fig. 3 PLFA biomarker concentrations in soil from Site A in August 2018, one year after the cessation of the N deposition treatment. Error bars are one standard error of the mean for three plots per treatment. Significant differences between the control and N deposition treatment for individual biomarkers are indicated by ** for $P < 0.05$ and * for $P < 0.10$

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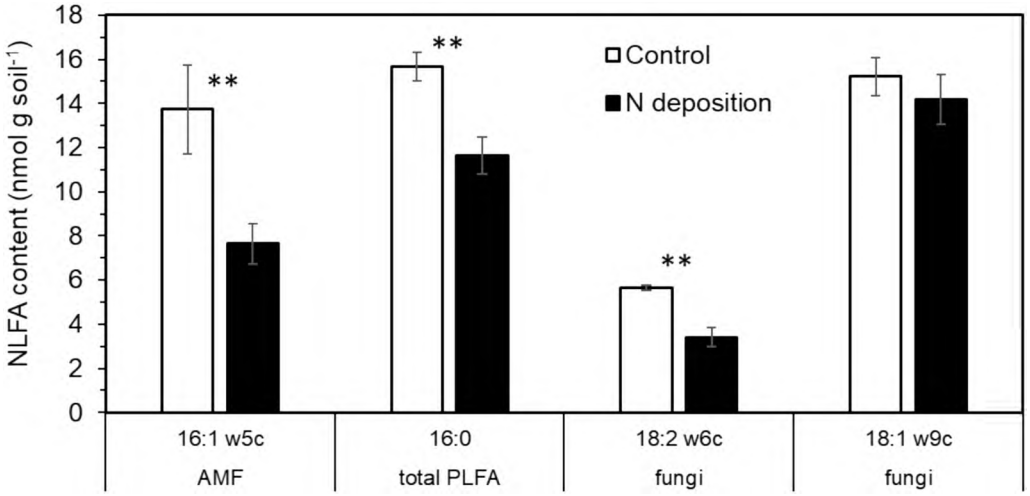


Fig. 4 NLFA biomarker concentrations in soil from Site A in August 2018, one year after the cessation of the N deposition treatment. Error bars are one standard error of the mean for three plots per treatment. Significant differences between the control and former N deposition treatment for individual biomarkers are indicated by ** for $P < 0.05$

Table 2 Effects of treatment, site, and their interaction on PLFA and NLFA contents in roots and soil from all four Michigan Gradient study sites three years after the cessation of N deposition treatments. Significant *P*-values are marked with asterisks (** for $P < 0.05$ and * for $P < 0.10$)

Substrate	Identification	Group	Treatment	Site	Interaction
Root–NLFA	16:1 ω5c	AMF	0.921	0.192	0.817
	18:1 ω 9c	Fungi	0.311	0.407	0.412
	16:0	Total1	0.386	0.427	0.879
Root–PLFA	18:1 ω 9c	Fungi	0.317	0.273	0.846
	15:0 iso	Gram Positive	0.822	0.131	0.243
	16:0	Total1	0.295	0.302	0.674
Soil–NLFA	16:1 ω5c	AMF	0.25	< 0.005**	0.123
	18:1 ω9c	Fungi	0.026**	0.158	0.553
	16:0	Total1	0.25	< 0.005**	0.408
	22:0	Total3	0.058*	0.005**	0.123
Soil–PLFA	16:1 ω5c	AMF	0.064*	< 0.005**	0.428
	19:1 (ω82) alcohol	Actinomycetes	0.201	0.966	0.492
	14:0	Bacteria1	0.306	< 0.005**	0.045**
	17:0 iso	Bacteria2	0.482	0.799	0.668
	17:0 anteiso	Bacteria3	0.044**	0.784	0.021**
	18:1 ω9c	Fungi	0.027**	0.120	0.355
	16:1 ω7c	Gram Negative1	0.108	0.013**	0.734
	17:0 cyclo	Gram Negative2	0.526	0.419	0.26
	15:0 iso	Gram Positive	0.027**	0.051*	0.080*
	16:0	Total1	0.527	< 0.005**	0.466
	18:0	Total2	0.483	< 0.005**	0.270
	22:0	Total3	0.556	< 0.005**	0.495

Discussion

All four sites studied in this project were N-saturated by the fifth to seventh year of the chronic N deposition treatment, as indicated by annual soil solution N losses increasing rapidly for several years after initiation of the experiment and then remaining consistent at levels

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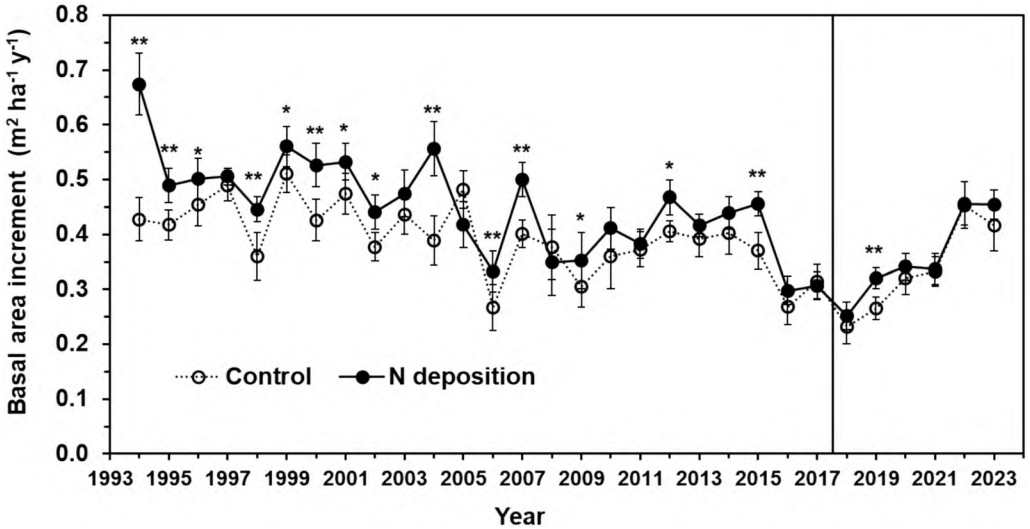


Fig. 5 Comparison of average basal area growth across all four sites during for the control and N treatments. Error bars are a standard error of the mean for all plots of a given treatment across the four study sites ($n = 12$). The vertical line indicates the end of the N deposition treatment after the 2017 growing season. Significant differences between the control and N deposition treatment for individual years are indicated by ** for $P < 0.05$ and * for $P < 0.10$

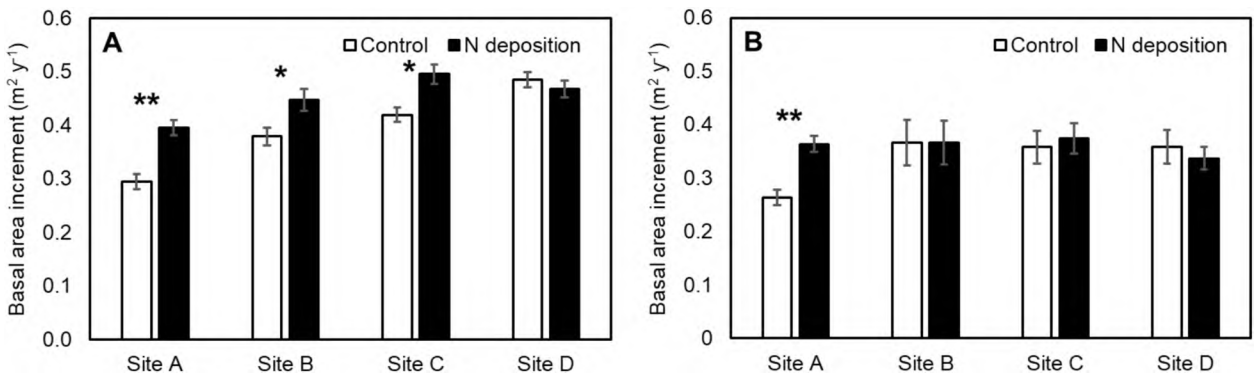


Fig. 6 Mean basal area increment by site and treatment for the period of active N deposition (**A**, 1994–2017) and the post treatment period (**B**, 2018–2023). Error bars are standard error of the mean for all plots and years of a given treatment within each site ($n = 72$ for **A** and $n = 18$ for **B**). Significant differences between the control and N deposition treatment within individual sites are indicated by ** for $P < 0.05$ and * for $P < 0.10$

approaching annual experimental N additions to treated plots (Pregitzer et al. 2004). In contrast, virtually no soil solution N loss occurred on control plots (Pregtizer et al. 2004). In addition, tree growth responded immediately to the experimental N additions, after which the amount of growth enhancement was fairly consistent, averaging 13% (Fig. 5). In concert with this, in 2001, after seven years of N addition, samples taken at the Michigan Gradient sites showed a decrease in total PLFA in soil in response to N additions and no difference in the proportion of microbial group indicators to each other (DeForest et al. 2004). Work done on the sites in 2005, 2006, and 2009 indicated that chronic N loading had a consistent negative effect on soil and root microbial groups, particularly for AMF (van Diepen et al. 2007, 2010, 2013; Zak 2015).

Microbial indicators

For microbes, potential responses to the cessation of experimental N deposition inputs could range anywhere from a rapid return to control plot community composition and function to a strong, continued suppression of microbial biomass and function. The latter possibility would be dependent on sufficiently enhanced site N availability continuing. Over the 24 years of treatment, each site received 72 g N m^{-2} of experimental N additions. Given this very large input, in combination with ambient

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N deposition rates across the region of approximately 0.5 to 0.9 g N m^{-2} y^{-1} from 2017 to 2019 (Table 1), we hypothesized that the state of N saturation induced at the sites by the long-term experimental N deposition (Pregitzer et al. 2004) could be maintained for several years, leading to continued broad suppression of microbial community biomarkers and enhancement of tree growth. Alternative scenarios included the opposite extreme of an almost immediate return to control conditions, as well as a variety of slower, gradual changes in microbial community composition, biomass, and function.

Our hypothesis was supported in the first year after N deposition treatments stopped by strong overall microbial suppression but was not well supported in year three (Table 3). By that time, the overall reduction in microbial biomarkers was much less (7% average decline vs. 28 to 29% decline in earlier years), with indicators for fewer microbial biomarkers continuing to show significant suppression. Biomarkers in soil did remain suppressed more than those of roots during the third year after the cessation of the N deposition treatment (Table 3) but to only about half the degree they were in the first year.

The reductions in differences among treatments for microbial biomarkers since the cessation of experimental N-loading are happening despite continuing moderate ambient N deposition (Table 1), which was not sufficient to maintain the effects of the long-term experimental N additions. Further recovery is likely over time due to cessation of N treatments. The more complete recovery of bacteria than fungal populations through three years may be due to the tendency for some fungal groups have increased sensitivity to N (Wang et al. 2021) and slower responses to environmental conditions.

Tree growth

Increased aboveground tree biomass growth coupled with decreased decomposition rates in response to chronic N addition can lead to greater C storage in anthropogenically N fertilized temperate forests. Higher above- and belowground C storage, at least temporarily, may improve forests' capacity as C sinks (Pregitzer et al. 2008). In the forests included in this experiment, enhanced growth due to the N deposition treatment declined over the first six years post-treatment and was no longer statistically significant. This occurred as biomarkers for AMF in roots and soil in the former N deposition treatment were no longer strongly suppressed relative to those for the control treatment. These findings are consistent with a mechanism by which an increase in AMF biomass following the end of experimental N additions led to greater C allocation to AMF, leaving less for other uses, including aboveground diameter growth.

Implications for ecosystem C sequestration and storage

Long-term forest growth measurements and model simulations both have suggested that elevated N deposition has contributed to increased tree growth and rates of forest C sequestration for several forest types (Li et al. 2025; Magnani et al. 2007; Quinn et al. 2010). As a result, impacts of elevated N deposition on tree growth are often included in ecosystem process models. Our findings suggest that such models should also be responsive

Table 3 Percent changes relative to the control for microbial biomarkers examined in the N deposition treatment plots in four northern hardwood forests in Michigan

Sample type	Microbial group	Treatment effect 2006 (%)	Treatment effect 2018 (%)	Treatment effect 2020 (%)
Root—PLFA	AMF	**−38**	**−35**	
	Fungi		**−24**	20
Root—NLFA	AMF	**−28**	**−60**	−7
	Fungi		−12	6
Soil—PLFA	AMF	**−41**	**−37**	**−22**
	Bacteria	**−22**	−20	−3
	Gram+ bacteria	−22	−17	**−21**
	Gram- bacteria	**−29**	**−32**	−9
	Actinomycetes	**−21**	−25	
	Fungi	**−27**	**−34**	**−23**
	Total	**−24**	**−25**	−4
Average Soil		**−27**	**−27**	−14
Average All		**−28**	**−29**	−7

Data from 2006 are from Van Diepen et al. (2010). Data for 2018 are for Site A, only. Values for microbial groups in a given year include all biomarkers for that group that were assessed in that year. Treatment effects in bold are statistically significant at $P < 0.05$

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to reduced N deposition, with reductions in N deposition resulting in lower tree growth and thus reduced rates of annual ecosystem C sequestration in just a few years. However, it should be noted that the C stored in living tree biomass during the period of higher N deposition should remain on site, first in living biomass and later in dead biomass as tree death occurs over time in association with stand dynamics.

The fairly rapid return of microbial biomarkers toward control levels also has implications for ecosystem C storage in surface mineral soil and forest floor. At the Michigan Gradient sites, multiple years of simulated elevated N deposition caused greater increases in C storage in these pools than occurred in living biomass (Pregitzer et al. 2008). The possibility exists that a recovering microbial community may promote greater decomposition rates, stopping the accrual of surface soil and forest floor C. In addition, it also may be possible for a recovering microbial community to use the extra C stored, causing gains that occurred under elevated N deposition to be lost. This appears to have occurred for the forest floor at our study sites, as Propson et al. (2024) found that elevated forest floor C stocks were gone by year five post-treatment. The elevated surface mineral soil C remained, but it is uncertain for how long, given the response in the forest floor. In concert with these responses, extracellular activities of peroxidase were greater for the former N deposition treatment than the control in the forest floor across all four sites but were still suppressed in the mineral soil (Propson et al. 2024).

In soil, several factors could allow the extra C stored during the long-term experimental N deposition to remain. First, much of the extra C stored in the surface soil was in the form of occluded particulate organic matter, which is considered to be physically protected from microbial access and thus has potentially long storage times (Zak et al. 2017). Also, it cannot be determined from our microbial biomarker data which genera are recovering. The structure of the microbial communities may be different at the species level now compared to pre-treatment due to differing N tolerances and needs (Strengbom et al. 2001). Thus, the apparent recovery of microbial populations indicated by the PLFA and NLFA biomarkers in mineral soil does not necessarily mean a full recovery of the services they once provided (Zak et al. 2019). The reduced extracellular peroxidase activity in the mineral soil at the sites (Propson et al. 2024) might be indicative of this, as not all fungal types produce this enzyme important in the degradation of lignin-like organic matter.

Beyond temperate deciduous forests, experiments on the recovery of soil microbial populations following reductions in elevated N inputs have been conducted in a boreal forest in Sweden and an alpine ecosystem in Colorado. These experiments showed limited to no recovery of mycorrhizal fungi, microbes, or other fungi nine years after the cessation of experimental N deposition (Bowman et al. 2018; Strengbom et al. 2001). In one of the longest N addition experiments in the literature, ectomycorrhizal fungi had not recovered in 23 years from 46 previous years of experimental N deposition in a forest in the Czech Republic, despite signs of lower nitrogen N in the recovering plots (Choma et al. 2017). Recovery of soil microorganisms can be slow, thus the level of recovery we observed in such a short time period was somewhat unexpected. This faster-than-expected recovery may have been due to favorable weather, more rapid processing of N than we originally thought, or atmospheric deposition being far too little to maintain chronic N addition effects (particularly in the northernmost site). As of yet, most long-term studies on impacts from anthropogenic N deposition focus on the impacts during active loading, and fewer have examined recovery rates. Future work may show this pattern of fairly rapid recovery is more common than previous studies have shown, especially in the temperature zone and/or in sites that were not severely N limited prior to receiving chronic elevated N inputs.

Given the strong suppression of soil microorganisms observed along the gradient during the N addition period of this study, we thought a slow recovery was reasonable to expect. Differences in soil type, forest type, growing season, and natural N deposition differentiate the Michigan Gradient study from studies conducted elsewhere, potentially contributing to the faster recovery rates found here. Importantly, the sites in this study were not extremely N limited to begin with, unlike sites studied elsewhere. Significant site effects and several statistically significant site by treatment interactions within the Michigan Gradient were found, suggesting that where along the gradient a site occurred influenced the microbial community detected therein. Overall, recovery of the microbial indicators at Site A often was less than for the other sites, and Site A is the site that had the strongest response to N additions (Figures S1 and 6, see also Propson et al. 2024 and Van Diepen et al. 2007, 2010, 2011), and likely was most N limited at the beginning of the study, due to inherently lower ambient N deposition over time. Site A's slower recovery of microbial biomarkers and continued growth enhancement (Fig. 6B) is consistent with the low to lack of recovery in the more N-limited ecosystems studied in Colorado and Sweden, where long-term treatments may have had a much greater impact on site N capital and cycling that continued post-treatment.

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Conclusions

Accurate modeling of C and N fluxes within ecosystems depends on accurate estimations of the response of those fluxes to changing environmental conditions, including rates of N deposition. Understanding how soil microorganisms react to increased N deposition in turn informs modeling of decomposition, N usage, net ecosystem production (NEP), net primary production (NPP), soil respiration, and C storage (Pregitzer et al. 2008). High levels of N deposition have been shown to have a dramatic effect on tree growth and soil carbon storage even over a relatively short time frame (Waldrop et al. 2004; Wang et al. 2021), but our results suggest that for sugar maple-dominated northern hardwood forests these responses may subside fairly quickly when N inputs are reduced. In places including eastern China and northern India, where N deposition has been very high over recent decades (Decina et al. 2020; Liu et al. 2022; Wen et al. 2020), our results suggests that ongoing impacts to their forests may lessen quickly if they are able to reduce N emissions, as appears to now be happening in parts of China (Liu et al. 2024). In the United States and western Europe where N deposition rates have decreased compared to peak industrial levels approximately four decades ago, results of this study suggest that microbially-mediated effects of N deposition on ecosystem carbon cycling and storage have likely been lessening for several decades.

Abbreviations

AMF	Arbuscular mycorrhizal fungi
ANOVA	Analysis of variance
ANPP	Aboveground net primary production
C	Carbon
dbh	Diameter at breast height
FAME	Fatty acid methyl ester
N	Nitrogen
NEP	Net ecosystem production
NLFA	Neutral lipid fatty acid
NPP	Net primary production
Oa	Organic a (sapric) horizon
Oe	Organic e (hemic) horizon
Oi	Organic i (fibric) horizon
PLFA	Phospholipid fatty acid

Supplementary Information

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Supplementary file 1.

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Author contributions

The project was conceptualized by Eileen Reeves and Andrew Burton with additional input from Katherine Higdon and Jennifer Eikenberry. Eileen Reeves, Katherine Higdon, and Jennifer Eikenberry collected and analyzed PLFA and NLFA samples and interpreted the data. Andrew Burton conducted annual tree measurements and analyzed tree growth data. The manuscript was written by Eileen Reeves, with feedback and editing from Andrew Burton and Katherine Higdon.

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Data availability

The datasets used and/or analyzed during the current study are available in Michigan Technological University's Digital Commons repository: https://digitalcommons.mtu.edu/all-datasets/74

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors have no competing interests.

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